higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N5

(8060045)

15 November 2019 (X-paper)
09:00–12:00

This question paper consists of 4 pages, 2 addenda and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL SURVEYING N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.1.5) in the ANSWER BOOK.

1.1 Isoheights are lines that join points of the same height above sea level.

1.2 The theorem of Pythagoras does not apply in the join calculation.

1.3 Stake value is another term that refers to a chainage.

1.4 Offsets are measured perpendicularly to the baseline.

1.5 Step-chaining is mainly used during the construction of a staircase.

(5 × 2) **[10]**

QUESTION 2

Define the following terms used in surveying:

2.1 Chaining

2.2 Direction

2.3 Cadastral surveying

2.4 Parallax

2.5 Booking

(5 × 2) **[10]**

QUESTION 3

3.1 A sloping rectangular site has been set out. As a site surveyor you are required to put profiles for excavation so as to level the site.

Explain how you would go about transferring your information levels onto the profiles, based on the length of your traveller. (10)

3.2 State TWO most common errors when reading a staff. (2 × 2) (4)

3.3 A square plot has an area of 25 m^2. If the land is to be represented on a plan of 1:200, find the length of a side in millimetres. (6)

3.4 Give TWO characteristics of contours according to the slope of the terrain. (2 × 2) (4)

[24]

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QUESTION 4

4.1 Explain how the two-peg method is used to adjust a tilting level to eliminate a collimation error. (9)

4.2 The TABLE on ADDENDUM A, shows notes taken during a levelling survey between point A and point B.

Reduce the readings using the rise and fall method. Do ALL the necessary checks for errors.

Complete the table, write your EXAMINATION NUMBER in the space provided on ADDENDUM A, detach it from the question paper and hand it in with the ANSWER BOOK. (20)
[29]

QUESTION 5

5.1 A line A–D was measured in three sections:

A–B 89,288 m at a slope of 2° 30'25"
B–C 100,020 m at a slope of 3° 40'30"
C–D 67 m at a slope of 1° 25'13"

Find the horizontal distance A to D. (10)

5.2 Calculate the slopes A–B, B–C, C–D and D–E of the given longitudinal section of the pipeline shown on ADDENDUM B. The slope must be given as a ratio. (17)
[27]

TOTAL: 100

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ADDENDUM A **EXAMINATION NUMBER:**

TABLE

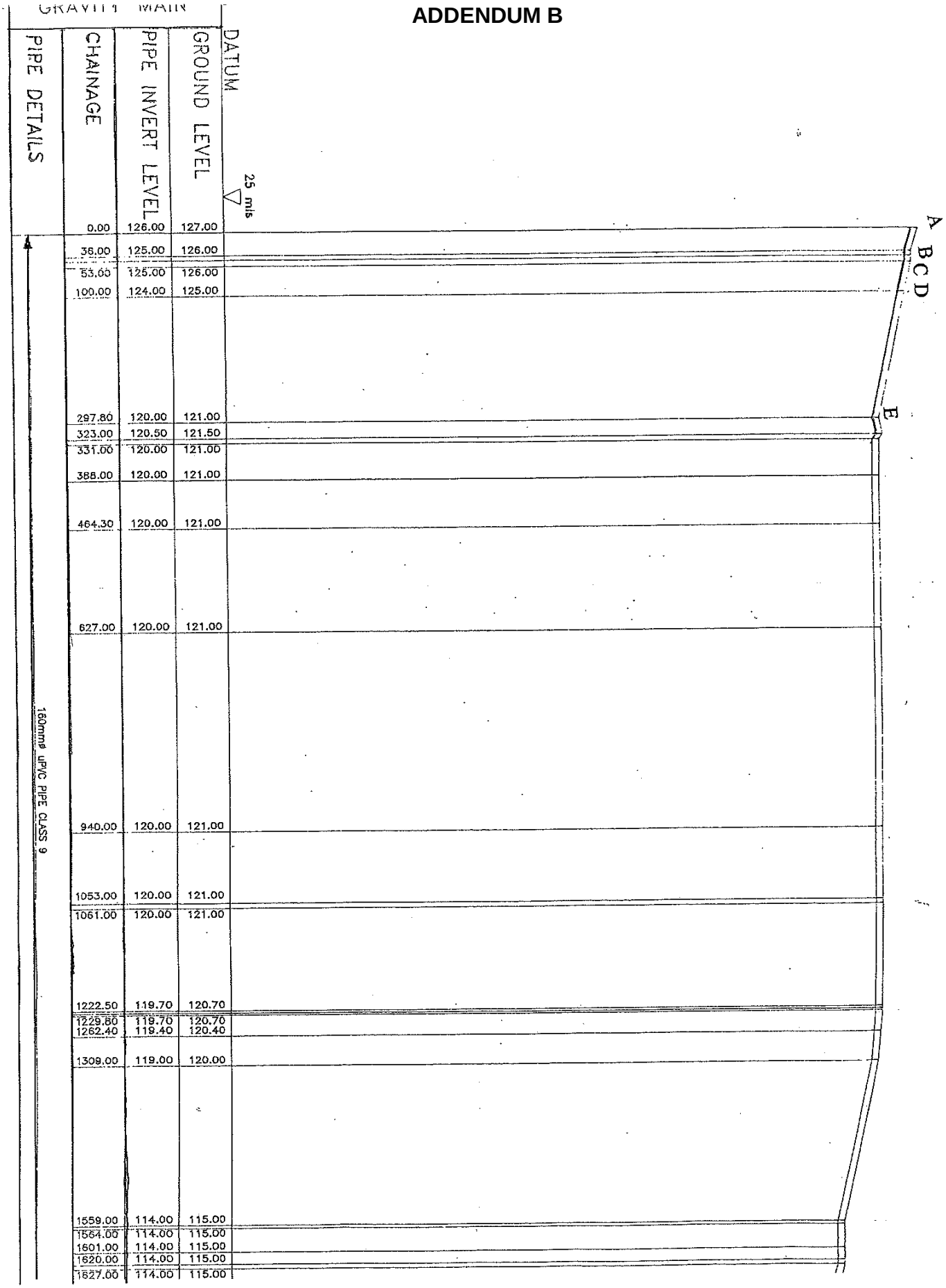
POINTS	BACK SIGHT	INTER SIGHT	FORE SIGHT	RISE	FALL	REDUCED LEVEL	REMARKS
A	0,49						
	0,27		3,29				
	0,39		3,77				
	3,72		3,59				
		1,11					
	3,56		0,82				
	3,89		1,36				BM 1025,00 above m.s.l.
	3,72		0,99				
	3,69		1,02				
	3,86		1,31				
	3,90		1,56				
B			2,40				

(20)

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ADDENDUM B



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FORMULA SHEET

Any applicable formula may be used.

$\Delta h = 50I\sin 2\Theta + HI - MH = 100I\sin\Theta\cos\Theta + HI - MH$

Or

$V = -KS\cos\Theta\sin\Theta$

$HD = 100/\cos^2\Theta$ *of* $KS\cos\Theta$

$Ct = L.e.(Tm-Ts)$, $Ct = L.e(Tm-Ts)$ *of* $L[1+e(Tm-Ts)]$

$Cs = L.(1-\cos\Theta)$

$Cs = H(\sec\Theta-1)$

$Ce = {}^{L.H}/_{R}$

$Slope = {}^{\Delta h}/_{HD}$

$V = {}^{d}/_{3}[(y_1+y_n) + 2(y_3+y_5+\ldots.+y_{n-2}) + 4(y_2+y_4+\ldots..+y_{n-1})]$

$\alpha = \tan^{-1}{}^{\Delta y}/_{\Delta x}$

$\alpha = \tan^{-1}{}^{\Delta x}/_{\Delta y}+90°$

$\alpha = \tan^{-1}{}^{\Delta y}/_{\Delta x}+180°$

$\alpha = \tan^{-1}{}^{\Delta x}/_{\Delta y}+270°$

$S = {}^{\Delta y}/_{\sin\alpha}$
$S = {}^{\Delta x}/_{\cos\alpha}$

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